Inferential Statistics: Sampling & Confidence Intervals

🔍 1. What is Inferential Statistics?

Inferential statistics helps us **make conclusions or predictions about a population** using a **sample** of data.

- **Population** = everyone or everything we want to know about
- **Sample** = a smaller group taken from the population

🧪 2. Sampling

✅ Why Do We Sample?

Studying an entire population is often **impossible** (too large, expensive, time-consuming). A **good sample** can represent the whole population.

🎯 Types of Sampling

Type of Sampling	Description	Example
Random Sampling	Every member has equal chance	Pick 100 students randomly from a school
Stratified Sampling	Population divided into groups (age, gender), and sample is taken from each	50 boys and 50 girls
Systematic Sampling	Select every k-th member	Choose every 10th customer
Convenience Sampling	Easy to reach people (risky, biased)	Survey your classmates only

> ⚠️ **Important:** A good sample must be **random & unbiased** to correctly represent the population.

📊 3. Sample Mean vs Population Mean

- **Population Mean (μ)** → True average of all data (unknown)
- **Sample Mean ($\bar{x}$)** → Average from our sample (known)

We use $\bar{x}$ to **estimate μ**.

🎯 4. Confidence Intervals (CI)

A **Confidence Interval** is a range of values that likely contains the true population mean.

Example: "*We are 95% confident that the average height of students is between 160 cm and 170 cm.*"

🧮 Basic Formula

```
Confidence Interval = x̄ ± Margin of Error
```

Margin of Error depends on:

- Sample size (n)
- Standard deviation (σ or s)
- Confidence level (90%, 95%, 99%)

🎚️ Common Confidence Levels

Confidence Level	Meaning
90%	Moderate certainty
95%	Most common in science
99%	Very high certainty

Higher confidence → Wider interval
Lower confidence → Narrower interval

🧭 Example (Simple)

A teacher takes a sample of **50 students**, average test score = **70**, margin of error = **±5**.

```
95% Confidence Interval = 70 ± 5 = [65, 75]
```

📌 **Interpretation:**
"We are 95% confident the true average score of ALL students is between **65 and 75**."

📢 Key Takeaways

Concept	Meaning
Sampling	Taking a small group to study a big population
Inferential Stats	Making predictions using sample data
Confidence Interval	Range that probably contains the true mean

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🧪 Practical in Python — Sampling & Confidence Intervals

🔧 1. Setup Libraries

```
import numpy as np
import pandas as pd
from scipy import stats
import matplotlib.pyplot as plt
```

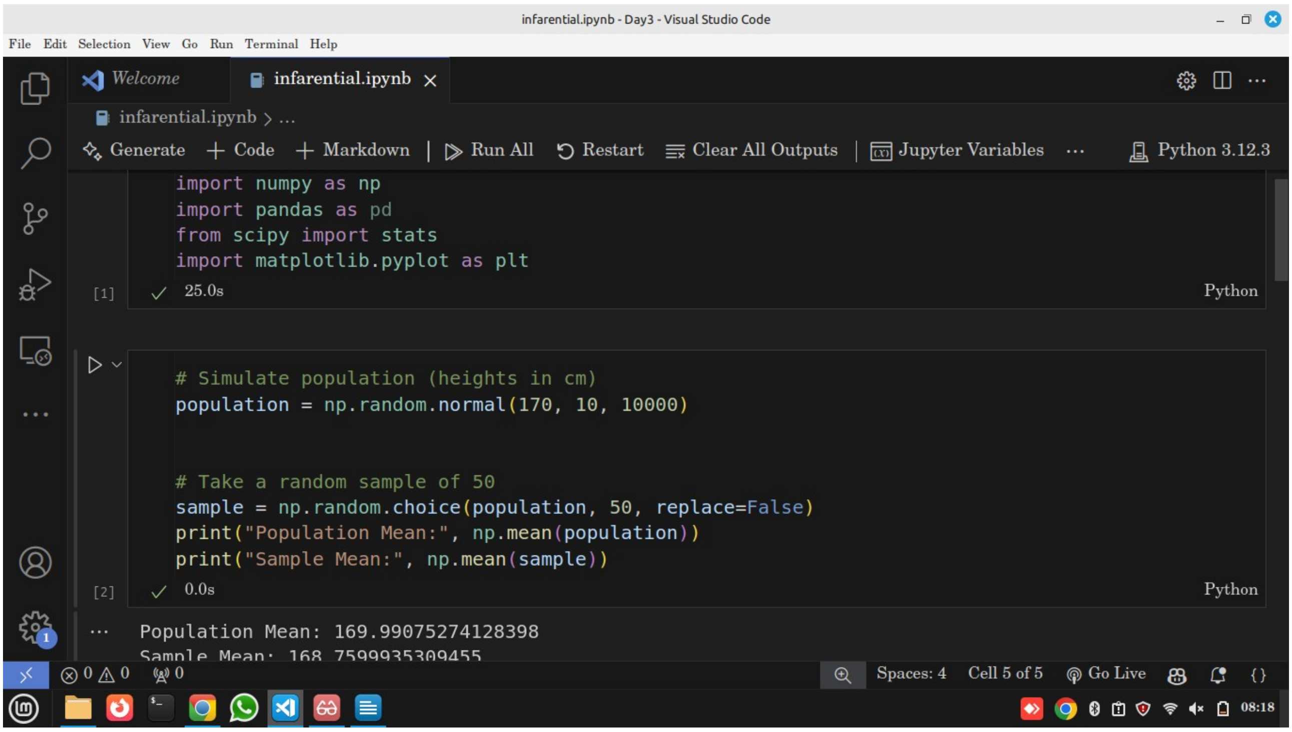
📖 Explanation

- `numpy (np)` → Used for generating numerical data (populations, samples, calculations).
- `pandas (pd)` → Usually used for data tables (not heavily used here, but often part of analysis).
- `scipy.stats` → Provides statistical functions like confidence intervals.
- `matplotlib.pyplot` → Used to visualize data and confidence intervals.

💡 Why Important?

These four libraries are the *core tools* for statistical analysis in Python.

🧠 2. Create a Population & Sample



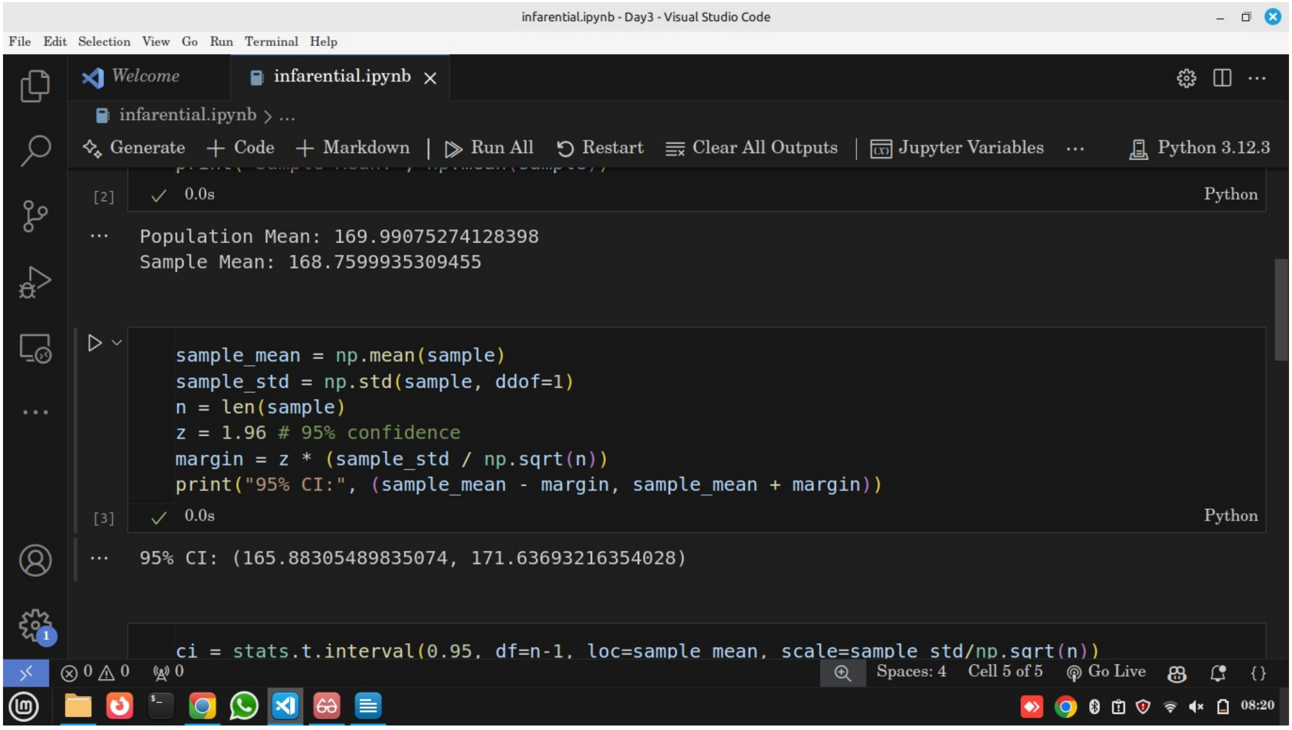
📖 **Explanation**

- `np.random.normal(170, 10, 10000)`
  → Creates **10,000 people's heights**, mean = 170 cm, std = 10 cm. (Population).
- `np.random.choice(..., 50)`
  → Randomly selects **50 people** from the population (Sample).
- We print both Population Mean and Sample Mean to compare real vs estimated value.

💡 **Insight (Real World)**

Just like surveys or opinion polls — we don't study everyone, we use a sample to estimate the population.

📏 3. Manual Confidence Interval (95%)

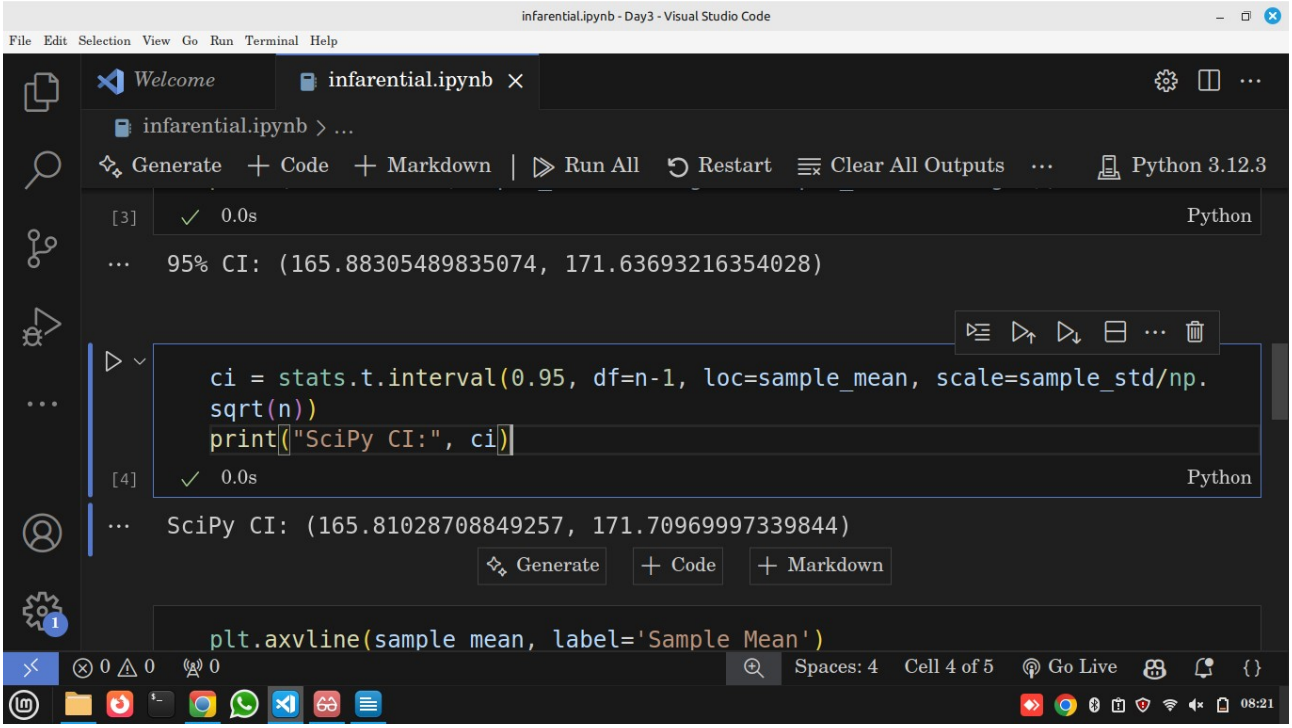


📖 **Explanation**

- `sample_mean` → Average height from our 50 people.
- `sample_std` → Variation within the sample (standard deviation).
- `n` → Sample size (50).
- `z = 1.96` → Z-score for **95% confidence**.
- `margin = z * (sample_std / sqrt(n))` → Formula for **margin of error**.
- Final CI = `sample_mean ± margin`.

💡 **Insight**
We are saying:
👉 *"We are 95% confident the true average height of the population lies within this range."*

🧪 4. SciPy Confidence Interval (Shortcut)

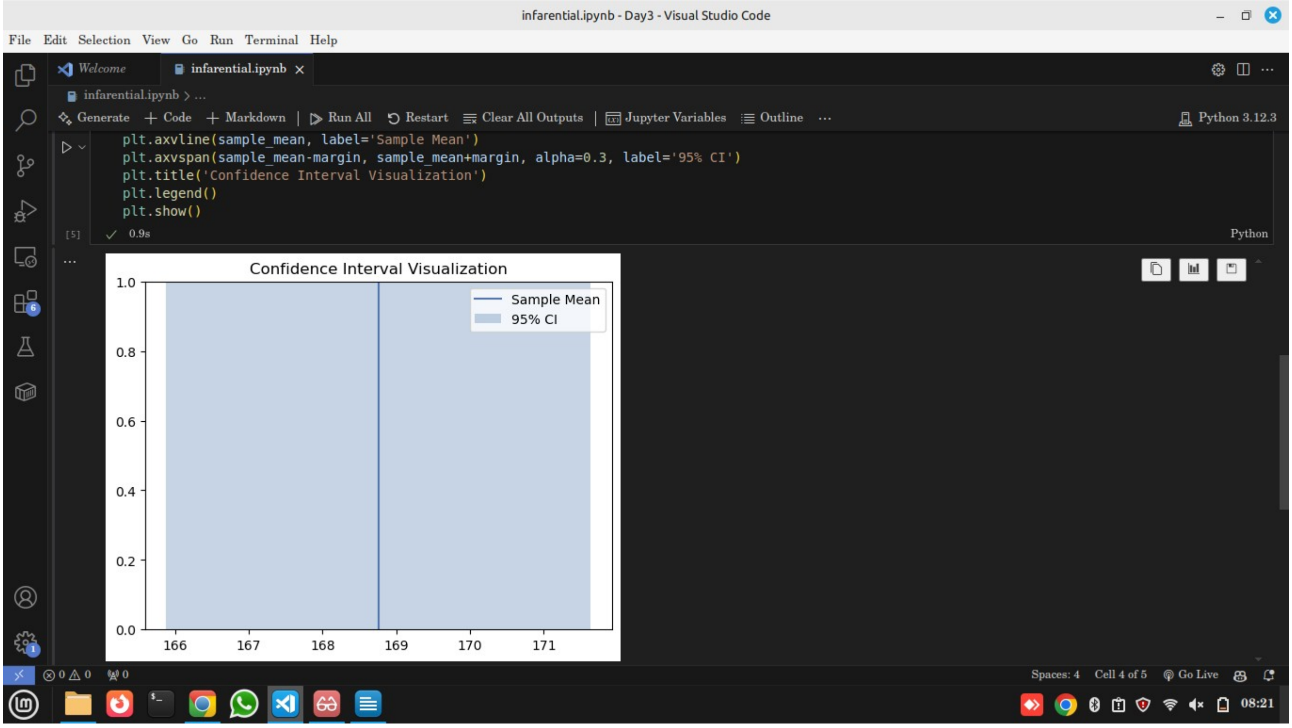


📖 **Explanation**

- `stats.t.interval` → Automatically calculates CI using the **t-distribution**.
- `0.95` → 95% confidence level.
- `df=n-1` → Degrees of freedom (49).
- `loc` → Sample mean.
- `scale` → Standard error (std / √n).

💡 **Why Use SciPy?**
You don't have to manually calculate — this is used in real data science projects for accuracy.

📊 5. Visualize Confidence Interval



📖 Explanation

- `axvline()` → Draws a vertical line at the sample mean.
- `axvspan()` → Shades the *confidence interval range*.
- `alpha=0.3` → Makes shading transparent.
- `legend()` → Shows labels.
- `show()` → Displays the plot.

💡 Insight

The shaded area shows all possible values where the **true population mean** could lie with 95% confidence.
If we repeat sampling 100 times → about 95 intervals will contain the true mean.

🌍 Real-World Applications of Sampling & Confidence Intervals

🏥 1. Medical Research & Drug Testing

When developing a new medicine, it's impossible to test **every single patient on Earth**.

✅ Researchers test a **sample** of patients
✅ Calculate average recovery rate
✅ Use **confidence intervals** to estimate how effective the drug is in the entire population

🗣️ Example:
"We are 95% confident that this drug improves recovery time by 4 to 6 days."

🗳️ 2. Opinion Polls & Elections

News agencies don't interview every citizen before an election.

✅ They survey a **sample** of voters
✅ Calculate percentage support for a candidate
✅ Use confidence intervals to predict national results

🗣️ Example:
"Candidate A is expected to get 52% ± 3% votes (95% CI → 49% to 55%)."

🏢 3. Customer Satisfaction & Market Research

Companies use surveys to test new products.

✅ They sample 1000 customers
✅ Estimate satisfaction level
✅ Use CI to infer how **millions** of customers might react

🗣️ Example:
"We are 95% confident that overall satisfaction is between 80% and 85%."

🛠️ 4. Manufacturing & Quality Control

Factories cannot test *every* product due to cost/time.

✅ They measure a sample of products (e.g. bottle weights)
✅ Use confidence intervals to estimate if machine is working correctly

🗣️ Example:
"Bottle weight is 500 ml ± 2 ml (95% CI)."

📈 5. Finance & Risk Estimation

Risk analysts compute average returns or losses using market samples.

✅ Use sample data (historical prices)
✅ Use CI to predict future earnings or risk range

🗣️ Example:
"Expected portfolio return is 8% ± 1.5% (95% CI)."

🧠 Why Data Scientists Must Learn This

Concept	Real Use
Sampling	Saves time and cost, works with real-world limits
Confidence Interval	Measures uncertainty and reliability
Estimate Population Mean	Vital for prediction & decision making

Assignment

🕒 Time: 15 Minutes

- **Short Answer & Practical Tasks**

📝 Part A – Conceptual (5 min)

Answer in 2–3 sentences each:

1. **Sampling Error:**
   What is sampling error, and why does it occur in statistics?
2. **Confidence Interval (CI):**
   What does a 95% confidence interval really mean?
3. **Sample vs Population:**
   Why is it often impractical to collect data from the entire population?

🔍 Part B – Practical Application (7 min)

A company wants to know the **average monthly spending** of its customers. A sample of **100 customers** shows an average spending of **$250** with a **standard deviation of $40**.

1️⃣ Calculate the 95% Confidence Interval

Use formula:
CI = Mean ± 1.96 × (SD / √n)

Show your steps.

- Mean = 250
- SD = 40
- n = 100

Questions:
a) What is the margin of error?
b) What is the confidence interval range?